

Corporal Punishment and Later Life Outcomes: Evidence from the Young Lives Study (Ethiopia, India, Peru and Vietnam)

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Executive Summary

Corporal punishment remains a prevalent practice in many low- and middle-income countries, despite growing evidence of its adverse effects and the introduction of legal prohibitions. This study informs the ongoing policy debate by examining the long-term impact of corporal punishment experienced at age 8 on cognitive and psychosocial outcomes at age 15, as well as educational, labor market, and mental health outcomes in early adulthood (ages 19-22).

The results show that corporal punishment is negatively associated with cognitive performance, with the most consistent and statistically significant effects found in Peru. Disaggregated results indicate negative associations for math scores in most countries except Ethiopia, and for vocabulary scores in Peru and India. For psychosocial outcomes, the most consistent effects are observed in Vietnam, where corporal punishment is negatively associated with overall psychosocial performance, agency, and self-esteem indices.

For early adulthood outcomes, there is limited evidence of an impact on mental health across countries. In Vietnam, corporal punishment is associated with a lower educational enrollment rate,¹ while in Peru and India it is associated with a higher employment rate.²

¹Measured as the average of three binary indicators of whether the respondent was enrolled in education at the time of each 2020 survey call.

²Measured as the average of three binary indicators of whether the respondent worked in the week before each 2020 survey call.

1. Introduction

Corporal punishment in schools remains one of the most pervasive forms of violence against children globally. Despite being legally prohibited in around two-thirds of countries, enforcement is weak and the gap between formal prohibition and classroom reality remains wide ([Global Initiative to End All Corporal Punishment of Children, 2020](#)). Surveys consistently find that physical punishment by teachers continues at high rates even in countries with explicit legal bans, suggesting that legislation alone has proven insufficient to change entrenched disciplinary norms and practices ([Global Initiative to End All Corporal Punishment of Children, 2020](#); [UN Committee on the Rights of the Child, 2006](#)). The scale of the problem is stark: among children aged 8 in the Young Lives study countries, over half in Peru and Vietnam, three quarters in Ethiopia, and more than nine in ten in India reported witnessing a teacher administer physical punishment in a single school week ([Oganda Portela and Pells, 2015](#)). Beyond prevalence, violence in school is among the foremost reasons children report disliking school, with over half of children in Vietnam and more than a quarter in India citing it as their primary concern ([Oganda Portela and Pells, 2015](#)) pointing to a direct channel through which punitive discipline may undermine children’s engagement with learning and willingness to attend.

The harms associated with corporal punishment extend well beyond the immediate physical act. A growing body of evidence links exposure to physical punishment to a wide range of cognitive and socio-emotional impacts, including impaired academic performance, reduced self-esteem, heightened anxiety and depression, and behavioral difficulties ([World Health Organization, 2025](#); [Hecker et al., 2014](#); [UN Committee on the Rights of the Child, 2006](#)). These associations have been observed across diverse country contexts and are consistent with broader theoretical frameworks linking adverse childhood experiences to disrupted development. Despite this broad picture of harm, the evidence base tracing a credible causal pathway between corporal punishment and children’s development outcomes remains limited. Most existing research is cross-sectional, making it impossible to disentangle whether punishment causes worse outcomes or whether children who are already struggling academically or behaviorally are more likely to be punished, a particularly acute concern given that teachers in several settings have been found to use physical punishment as a direct response to poor academic performance ([Baker-Henningham et al., 2009](#)). Studies

from high-income countries, principally the United States, dominate the literature ([Gershoff, 2002](#); [Paolucci and Violato, 2004](#)), limiting the generalizability of findings to the low- and middle-income settings where corporal punishment in schools is most prevalent. Where longitudinal data have been used to partially address reverse causation, follow-up periods have typically been short, leaving open the question of whether any documented effects persist into adolescence and adulthood or fade as children age out of punitive school environments. This project addresses these gaps directly. It builds on the work of ([Oganda Portela and Pells, 2015](#)), who used Young Lives longitudinal data from Ethiopia, India, Peru, and Vietnam to document significant negative associations between corporal punishment at age 8 and cognitive and psychosocial outcomes at ages 8 and 12. Their paper had direct policy influence, contributing to advocacy that resulted in new legal restrictions on corporal punishment in Peru. The Young Lives study has since collected two additional survey rounds, with respondents aged approximately 15 at Round 5 and 22 at Round 6. These rounds include measures of mental health, education enrollment, and employment, making it possible for the first time to trace whether early childhood exposure to punitive discipline carries consequences well into young adulthood.

The core research question is whether experiencing corporal punishment at age 8 shapes outcomes at ages 15 to 22. The analysis employs ordinary least squares as the baseline estimator, supplemented by cluster fixed effects to account for unobserved cluster-level confounders, and an instrumental variable strategy using the leave-one-out experience of classmates to strengthen claims about the direction of causality. The richness of the Young Lives data, which includes test scores and family background characteristics measured prior to the corporal punishment reports, allows for unusually strong controls relative to most observational studies in this literature. Four countries are included, Ethiopia, India, Peru, and Vietnam, providing a diverse empirical base that varies considerably in the prevalence of corporal punishment, legal context, and economic development, and allowing for examination of both cross-country patterns and country-specific findings.

2. Literature Review

A large and growing body of research highlights the central role of early-life environments in shaping both cognitive and non-cognitive skills, with long-term implications for health, economic and social outcomes. As emphasized by [Heckman \(2011\)](#), investments in early childhood yield high returns when they foster supportive developmental environments. Conversely, adverse experiences, such as corporal punishment, may disrupt this trajectory. Corporal punishment is defined as any punishment in which physical force is used with the intention of causing some degree of pain or discomfort ([UN Committee on the Rights of the Child, 2006](#)).

2.1 Consequences of Corporal Punishment in Schools

This section primarily summarizes empirical findings on the effects of corporal punishment in schools, drawing on both cross-sectional and longitudinal studies, with a focus on low- and middle-income countries.

2.1.1 Cognitive and Academic Consequences

The empirical literature overwhelmingly documents a negative association between corporal punishment and children’s academic performance, although causal identification remains challenging. Cross-sectional evidence from Jamaica ([Baker-Henningham et al., 2009](#)) and Pakistan ([Naz et al., 2011](#)) shows that children exposed to higher levels of school-based violence perform worse in core subjects and exhibit higher dropout rates.

More rigorous longitudinal studies strengthen this conclusion. Using Round 3 and Round 4 of India’s Young Lives data, [Maiti \(2021\)](#) finds that exposure to school corporal punishment had a statistically significant negative impact on literacy and mathematics test scores, with effects persisting over time. Similarly, [Oganda Portela and Pells \(2015\)](#), using the same Young Lives dataset across Ethiopia, India, Peru and Vietnam, shows that experiencing corporal punishment at age 8 is negatively associated with concurrent mathematics scores across all four countries. Considering later impact, exposure to corporal punishment at age 8 was negatively associated with mathematics scores measured at age 12 in India, Peru and Vietnam.

Nevertheless, some studies report mixed or counterintuitive findings. Evidence from Malawi ([Psaki et al., 2017](#)) shows that school-related physical violence is associated with both lower absenteeism and better subsequent academic performance for females. Similarly, a cross-

sectional study in Uganda finds that past-week physical violence was associated with a detrimental educational outcome for females, while the relationship was reversed for males.

However, these findings should be interpreted with caution, as measuring the causal impact of violence on learning remains inherently challenging (Evans and Hares, 2023). Selection bias may partly explain why some studies report unexpected or counterintuitive relationships. For instance, schools that actively monitor and report incidents of violence may also be better managed and more academically effective. Taken together, the weight of the evidence consistently suggests that corporal punishment is detrimental to children’s learning and development.

2.1.2 Psychosocial Consequences

The longitudinal Young Lives research indicates that early exposure to corporal punishment is associated with lower self-efficacy, reduced agency, and higher feelings of shame (Oganda Portela and Pells, 2015). Similarly, the study in Malawi (Psaki et al., 2017) reports that physical violence is associated with poor mental health.

These findings are consistent with broader international evidence. Drawing on meta-analytic and longitudinal research across high-, middle-, and low-income countries, Smith (2006) concludes that corporal punishment is associated with higher levels of aggression, anxiety, and antisocial behavior, weaker parent-child relationships, and poorer mental health.

More recent evidence deepens this picture. Durrant and Ensom (2020) identify a strong link between corporal punishment and mental health problems such as anxiety, depression, low self-esteem, substance abuse, and suicidal tendencies. Ferrara et al. (2019) corroborates this connection, noting that poor mental health often manifests as fear of expression, heightened caution, and even increased aggression, which creates a cycle in which victims may become perpetrators of violence.

Similarly, Bangash et al. (2024) highlight that physical punishment erodes self-confidence, and increases anxiety and suicidal ideation. Zhang et al. (2025) further shows that these psychological harms translate into lower academic achievement and long-term developmental setbacks. Ultimately, the lack of a safe and supportive school environment undermines both emotional well-being and learning.

2.2 Drivers of Corporal Punishment

Understanding the determinants of corporal punishment is essential for interpreting its observed effects. At the individual level, younger children face higher rates of corporal punishment during their early years of schooling compared to older students (Oganda Portela and Pells, 2015). The same study finds that boys are more likely to report experiencing corporal punishment at all ages than girls. Socioeconomic status and geographic location also matter, although patterns vary across contexts. In India and Peru, children from poorer households and those living in rural areas are most likely to report being physically punished, whereas in Ethiopia and Vietnam, the reverse is observed: children in urban or relatively wealthier households report higher rates.

Teacher and staff characteristics are also central drivers of corporal punishment in LMICs. A qualitative and survey-based study in Sri Lanka indicates that teachers view corporal punishment as a legitimate disciplinary tool, particularly in overcrowded classrooms (Lakshman, 2018). This finding aligns with evidence from Uganda, which identifies teacher stress and the absence of alternative disciplinary measures as key determinants of corporal punishment (Ssenyonga et al., 2019). Another study in Uganda finds that teaching and non-teaching staff who have experienced or perpetrated intimate-partner violence, or who engage in violence outside the school, are more likely to punish students physically (Merrill et al., 2017).

Family characteristics further shape children’s exposure to physical violence. Children whose caregivers have lower levels of education are more likely to be subjected to such punishment, and students from low-income families are punished for reasons linked to poverty, such as lacking proper uniforms or being unable to pay fees (Oganda Portela and Pells, 2015). These family-level characteristics often overlap with school- and individual-level factors, such as attending under-resourced public schools or living in rural areas. Evidence from Nigeria shows that children from low-income families are more likely to attend schools where violence is more prevalent (Elendu et al., 2025). Caregivers’ acceptance of physical discipline at home further contributes to the persistence of corporal punishment in schools, as parents who view it as legitimate are less likely to advocate for non-violent alternatives (Schulte, 2021).

Finally, social norms play a key role in shaping teachers’ use of corporal punishment. Some scholars trace the origins of corporal punishment to traditional African societies that used physical discipline to

enforce communal values and hierarchical order (Elendu et al., 2025). Religious teachings and sociocultural beliefs have also historically justified corporal punishment, and these descriptive norms continue to underpin its persistence. In Sri Lanka, for example, teachers view acts of physical punishment as moral guidance rather than cruelty, necessary for “building character and teaching respect” (Lakshman, 2018). Similarly, in Tanzania, corporal punishment is regarded as an “act of care rather than hostility” (Zuberi et al., 2025). In Uganda, teachers are viewed as authority figures who have the right to punish in order to deter misbehavior (Ssenyonga et al., 2019). In Nigeria, the widespread acceptance of corporal punishment as a legitimate disciplinary tool has likewise contributed to its persistence (Elendu et al., 2025). The persistence of corporal punishment is further reinforced through the intergenerational transmission of descriptive norms. Teachers’ own experiences of being punished as children often inform their belief that corporal punishment is a necessary disciplinary tool (Lokot et al., 2020).

2.3 Legal and Policy Context of Corporal Punishment

Globally, efforts to prohibit corporal punishment in educational settings have accelerated. Of the 223 states and territories tracked, 106 have enacted laws banning corporal punishment in all schools (Pinheiro, 2006). However, the domestic translation of these prohibitions into effective measures and practices remains uneven. This section examines the country context of legal frameworks governing corporal punishment in schools, focusing on the year of Young Lives Round 3 (2009), when exposure to corporal punishment is measured.

2.3.1 Ethiopia

Ethiopia’s legal framework in 2009 did not explicitly prohibit corporal punishment across all settings. The Criminal Code of 2005 prohibits acts that cause bodily injury but allows for “disciplinary measures” by parents, guardians, and teachers, provided they are “within the law” (Federal Democratic Republic of Ethiopia, 2005). Similarly, the Civil Code of 1960 recognizes the right of parents to correct their children “reasonably,” creating a broad exemption that effectively legitimizes physical punishment in both homes and schools (Global Initiative to End All Corporal Punishment of Children, 2010). However, corporal punishment was not included among the disciplinary measures permitted under school administration regulations issued by the Ministry of

Education in 1998 ([Global Initiative to End All Corporal Punishment of Children, 2014a](#)). Therefore, while broader legal provisions allowed reasonable discipline, school-level regulations did not formally authorize corporal punishment, resulting in a degree of legal ambiguity.

2.3.2 India

By 2009, India had moved toward formal prohibition of corporal punishment in schools. The Right of Children to Free and Compulsory Education Act (2009) explicitly bans physical punishment and mental harassment in educational institutions ([Government of India, 2009](#)). Despite the formal ban in schools, Young Lives data from Round 3 (2009) indicate that nearly all children reported witnessing corporal punishment in classrooms, highlighting a substantial gap between legislation and practice ([Young Lives, 2015](#)).

2.3.3 Peru

While the passage of Law No. 30403 in December 2015 made Peru one of the first countries in the region to explicitly prohibit all forms of physical and humiliating punishment of children and adolescents, such a comprehensive legal framework was not yet in place in 2009. At that time, Peru relied on general child protection principles and normative discouragement within the education system, including Supreme Decree No. 007-2001-ED and the Code of Children and Adolescents (2000). Consistent with this, [Young Lives \(2015\)](#) indicate that more than half of children reported witnessing corporal punishment in schools.

2.3.4 Vietnam

Corporal punishment was formally prohibited in schools, as reflected in policy documents issued by the Ministry of Education and Training. The Vietnamese Education Law (2005) states that teachers must not disrespect the honour and dignity of learners, or physically harm or abuse them ([Global Initiative to End All Corporal Punishment of Children, 2014b](#)), and Decision 16/2008 further reinforces professional ethical standards for teachers ([Ministry of Education and Training of Vietnam, 2008](#)). Despite these formal restrictions, more than half of children reported exposure to corporal punishment in schools ([Young Lives, 2015](#)).

3. Data

This study uses data from Rounds 2-3 and 5-6 of the Young Lives Survey, a longitudinal study that has followed children in Ethiopia, India, Peru and Vietnam since 2002. Its panel structure, detailed household and child-level measures, and consistent survey design across countries make it well suited to show how early life experiences, such as exposure to corporal punishment, shape child and young adult development. This study focuses on the younger cohort, born in 2001-2002. Table 1 shows the survey rounds used in this study: Round 2 and 3 serve as the baseline data, while rounds 5 and 6 capture later life outcomes.

Table 1
Overview of Young Lives Survey Rounds Used

Round	Year	Age	Focus
2	2006–2007	5	Early schooling participation, cognitive development, nutritional status
3	2009–2010	8	Primary schooling, household educational investments, children’s time allocation, corporal punishment
5	2016–2017	15	Secondary school transitions, academic achievement, safety and emotional well-being
6	2022–2023	19–22	Early adulthood, tertiary education participation, labor market entry, household formation, agency and aspirations

3.1 Previous Research

This study builds on prior work by [Oganda Portela and Pells \(2015\)](#), who examine the relationship between corporal punishment and school outcomes using Young Lives data from Ethiopia, India, Peru, and Vietnam. They find that exposure to corporal punishment is negatively associated with concurrent outcomes at age 8 (Rounds 3) and later outcomes at age 12 (Round 4).

This paper extends that analysis by investigating whether corporal punishment has an impact on later life outcomes (ages 15-22). The longitudinal structure of the data allows us to assess whether early exposure to corporal punishment has lasting effects on developmental outcomes, or if its influence diminishes over time. To ensure comparability with the existing literature, we closely follow [Oganda Portela and Pells \(2015\)](#) in the construction of key independent and dependent variables, adapting them as needed to reflect the later survey rounds.

3.2 Variable Description

Table 2 presents the main outcome variables used in the analysis, grouped into four categories: cognitive outcomes, psychosocial outcomes, mental health, and educational enrollment and employment. These outcomes capture the main channels through which corporal punishment may affect child development.

Table 2
Outcome Variables

Indicator/Question	Data Type
Cognitive outcomes (Round 5)	
Cognitive performance	Numerical
Maths score, standardised	Numerical
Vocabulary score (PPVT), standardised	Numerical
Psychosocial outcomes (Round 5)	
Psychosocial composite index	Numerical
Agency index	Numerical
Shame index	Numerical
General self-efficacy	Numerical
General self-esteem	Numerical
Mental Health (Round 6)	
Generalized Anxiety Disorder Assessment (GAD-7)	Numerical
Patient Health Questionnaire (PHQ-8)	Numerical
Educational enrollment and employment (Round 6)	
Participant currently enrolled in education	Binary
Worked in the week before call	Binary

The cognitive performance is constructed as the average of standardized math and vocabulary scores, and is then standardized to have mean zero and standard deviation one. The psychosocial composite index is constructed following ([Anderson, 2008](#)) and combines four sub-indices: agency, shame, self-efficacy, and self-esteem. Each sub-index is standardized to have mean zero and standard deviation one, with variables recoded so that higher values indicate better outcomes (with the exception of the shame index, which is reversed). The sub-indices are aggregated using inverse-covariance weighting, which ensures a common direction across components and assigns greater weight to those that are less correlated with others.

Appendix A6 provides the detailed questions used to construct each sub-index. The agency index draws on items measuring perceived control, decision-making ability, and confidence in shaping one's own future. The shame index captures children's emotional responses to situations evoking feelings of inadequacy, vulnerability, and fear of judgment. General self-efficacy measures the extent to which a child believes they can handle difficulties and achieve goals through their own effort. General self-esteem captures a child's overall sense of self-worth, reflecting how positively they view themselves in terms of ability and value.

As visible in Table 3, the key independent variables are outlined, beginning with the main variables of interest: experience of corporal punishment and witness of corporal punishment. These are ordinal and categorical variables that capture whether and how frequently children are exposed to corporal punishment. The table also includes a set of control variables, capturing key child-level characteristics such as age, gender, location, and anthropometric indicators (weight-for-age and height-for-age z-scores). In addition, household-level characteristics are included such as total expenditure, household size, a wealth index, and the educational attainment of the caregiver, mother and father.

Table 3
Independent Variables

Category	Indicator/Question	Data Type
Corporal punishment (Round 3)	Experiencing corporal punishment	Categorical (ordinal)
	Witnessing corporal punishment on others	Categorical (ordinal)
Other controls (Round 2-3)	Age in months	Numerical
	Gender	Categorical (nominal)
	Location	Categorical (nominal)
	Weight-for-age z-score	Numerical
	Height-for-age z-score	Numerical
	Household expenditure	Numerical
	Household size	Numerical
	Wealth index	Numerical
	Maths score, standardised (2009)	Numerical
	Agency index (2009)	Numerical
	Shame index (2009)	Numerical
	CDA score, raw score (2006)	Numerical
	PPVT score, standardised(2006)	Numerical
	Caregiver's education (2006)	Categorical (ordinal)
	Mother education (2006)	Categorical (ordinal)
	Father education (2006)	Categorical (ordinal)

3.3 Attrition

Table 4 reports the number of children successfully re-interviewed across survey rounds for the four Young Lives countries. Overall, attrition between Round 3 and Round 5 is modest. Vietnam and India exhibit the lowest attrition rates (1.81 percent and 1.53 percent, respectively), while Peru exhibits the highest at 5 percent. Attrition between Round 3 and Round 6 is higher, reflecting that Round 6 was conducted during the COVID-19 and data were collected via telephone survey.

Table 4
Sample Attrition by Country and Round

Country	Round 3	Round 5		Round 6	
	Sample Size	Not Attrited	Attrited	Not Attrited	Attrited
Vietnam	1882 100%	1848 98.19%	34 1.81%	1667 88.58%	215 11.42%
Ethiopia	1445 100%	1410 97.58%	35 2.42%	1295 89.62%	150 10.38%
India	1901 100%	1872 98.47%	29 1.53%	1841 96.84%	60 3.16%
Peru	1898 100%	1803 95.00%	95 5.00%	1725 90.88%	173 9.12%

3.3.1 Balance Tests for Attrition Bias by Country

Table 5 reports differences in means between children who remained in the sample and those who attrited, across all four countries and both rounds. Attrition is not entirely random. In Round 5, statistically significant differences are observed in Ethiopia, where attritors scored higher in math and reported higher shame indices, and in India, where attritors had higher agency scores. In Round 6, attrition becomes more systematic: in Peru, attritors scored significantly lower in both math and vocabulary, while in Vietnam, attriting children were more likely to be male and exhibited lower math scores and agency index at baseline. Tables A1, A2, A3, and A4 provide a breakdown of attrition patterns on a country-by-country basis. These patterns suggest that attrition may be correlated with both cognitive ability and psychosocial characteristics, which could introduce bias into the estimates.

Table 5
Balance Tests: Difference in Means

Variable	Round 5				Round 6			
	Peru	Vietnam	Ethiopia	India	Peru	Vietnam	Ethiopia	India
Math Score (Std.)	0.087	-0.232	-0.405**	-0.236	2.223***	0.161**	-0.017	0.107
Vocabulary Score (Std.)	-0.173	-0.200	-8.809	-0.144	4.979***	0.079	1.128	-0.161
Agency Index	-0.052	0.043	-0.072	0.368**	0.178**	0.187**	-0.102	-0.012
Shame Index	-0.032	0.040	-0.363**	-0.045	-0.095	0.150*	0.038	-0.056
Male	-0.002	0.010	0.062	0.125	-0.063	-0.102***	0.020	-0.014
Corporal Punishment	-0.046	-0.129*	0.037	0.084	-0.031	-0.051*	0.013	0.126**

* p<0.1, ** p<0.05, *** p<0.01.

3.3.2 Inverse Probability Reweighting

Given the non-random attrition, we use Inverse Probability Weighting (IPW) to mitigate potential bias in the estimates. In order to control for this, we reweighted the sample prior to running any analysis.

First, the probability of attrition is estimated using a logistic regression model, where the binary attrition indicator is regressed on a set of baseline covariates. These covariates are: mathematics score, vocab score, agency index, shame index, corporal punishment experience, and gender. They were selected on the basis that the attrition analysis identified them as systematically related to sample dropout.

Second, the predicted probability of remaining in the sample is derived by taking the complement of the estimated attrition probability. Each observation is then assigned a weight equal to the inverse of this probability, such that children who resembled those more likely to attrite are upweighted in the analysis, restoring representativeness to the estimation sample.

Third, to prevent extreme weights from unduly influencing the results, weights are trimmed at the 1st and 99th percentiles. Prior to applying IPW, a missing covariate diagnostic confirmed that 11% of observations had incomplete baseline data, and that missingness was not strongly concentrated among attritors, limiting the scope for additional missing data bias.

4. Empirical Strategy

This study employs a three-part empirical strategy to understand the impact of having been exposed to corporal punishment at age 8 on subsequent developmental outcomes.

4.1 Post-Double-Selection LASSO

We use the post-double-selection LASSO approach (Belloni et al., 2014) to select control variables and mitigate omitted variable bias. The LASSO objective function minimizes the residual sum of squares subject to a penalty on the size of the coefficients, thereby reducing overfitting.

Following standard practice, we first regress the outcomes on the full set of candidate controls. Second, we regress the exposure to corporal punishment on the same set of controls. Finally, we take the union of variables selected in both steps as the final control set for the subsequent regressions.

4.2 Ordinary Least Squares (OLS) Regressions

We begin our main empirical analysis by examining the association between exposure to corporal punishment measured in Round 3 ($t = 3$) and developmental outcomes measured in Rounds 5 and 6 using Ordinary Least Squares (OLS) regressions, as expressed in Equation 1:

$$Outcome_{i,t} = \beta_0 + \beta_1 \cdot CP_{i,3} + \gamma \cdot X_{i,3} + \varepsilon_{i,t} \quad (1)$$

Here, $X_{i,3}$ is a vector of control variables selected by PDS LASSO, measured at or before Round 3, and $CP_{i,3}$ denotes reported exposure to corporal punishment at age 8 (Round 3). The coefficient β_1 represents the association between corporal punishment exposure and later outcomes, holding observed covariates constant.

The subsequent outcomes ($Outcome_{i,t}$) in this analysis cover multiple domains of child and young adult development. In Round 5, we examine both cognitive and psychosocial outcomes, while in Round 6 we focus on mental health and well-being, educational enrollment, and employment. The psychosocial outcomes in Round 5 comprise four indicators: agency index, shame index, general self-efficacy, and general self-esteem. Given the overlap and correlation among these measures, we construct a composite psychosocial index following Anderson (2008).

Additionally, we extend the specification to include cluster fixed effects (μ_c), which control for unobserved characteristics shared by children within the same geographical area, as expressed in Equation (2):

$$\text{Outcome}_{i,c,t} = \beta_0 + \beta_1 \cdot CP_{i,c,3} + \gamma \cdot X_{i,c,3} + \mu_c + \varepsilon_{i,c,t} \quad (2)$$

Lastly, we extend the cluster fixed effects specification to allow for heterogeneous effects across relevant dimensions, including gender, urban locality, and caregiver education, as illustrated in Equation (3):

$$\text{Outcome}_{i,c,t} = \beta_0 + \beta_1 \cdot CP_{i,3} + \beta_2 \cdot H_i + \beta_3 (CP \cdot H_i) + \gamma \cdot X_{i,c,3} + \mu_c + \varepsilon_{i,c,t} \quad (3)$$

H_i denotes the dimension of interest. While β_1 captures the association between corporal punishment and outcomes for the reference group, β_3 captures the differential effect for subgroup $H_i = 1$. A statistically significant β_3 would indicate that the relationship between corporal punishment exposure and developmental outcomes differs across the examined subgroups.

4.3 Instrumental Variables (IV) Estimation

To address potential endogeneity and obtain causal estimates, we employ an instrumental variables (IV) approach. Specifically, we implement a leave-one-out strategy, using the average level of corporal punishment witnessed by other children in the same community as an instrument ($Z_{s,3}$) for the child's own exposure.

Formally, the first-stage and second-stage equations are given by:

$$CP_{i,s,3} = \alpha_0 + \alpha_1 Z_{s,3} + \delta X_{i,s,3} + \nu_{i,s,3} \quad (4)$$

$$\text{Outcome}_{i,s,t} = \beta_0 + \beta_1 CP_{i,s,3} + \gamma X_{i,s,3} + \eta_{i,s,t} \quad (5)$$

The first stage (Equation 4) predicts individual exposure to corporal punishment based on the surrounding disciplinary environment. The second stage (Equation 5) estimates the impact of the predicted exposure on subsequent outcomes.

5. Results

This section is organized into four subsections. We first document descriptive patterns of corporal punishment exposure across the four countries, then present baseline OLS estimates alongside specifications with cluster fixed effects, followed by instrumental variables (IV) estimates. The section concludes with a heterogeneity analysis examining whether the estimates differ across relevant subgroups.

5.1 Exposure to Corporal Punishment

The descriptive statistics show substantial variation in exposure to corporal punishment across four countries. As shown in Figure 1, children are more likely to witness corporal punishment than to experience it directly. This difference is particularly large in Ethiopia, where 76.4% of children report witnessing corporal punishment, compared to 37.9% who report experiencing it, suggesting that a large share of children are exposed to a climate of classroom violence without being direct targets themselves. A similar pattern holds in India, where 92.3% of children witness corporal punishment while 77.2% report experiencing it, the highest rates of any country in the sample. In contrast, Peru and Vietnam exhibit lower prevalence on both measures. These patterns suggest that many children are exposed to corporal punishment indirectly through observing peers, even if they are not directly punished. Such indirect exposure may also influence school experiences, cognitive and non-cognitive outcomes.

In order to get a better understanding of the differing corporal punishment exposure rates, the figures below shows the rates disaggregated by gender, location and parental education level.

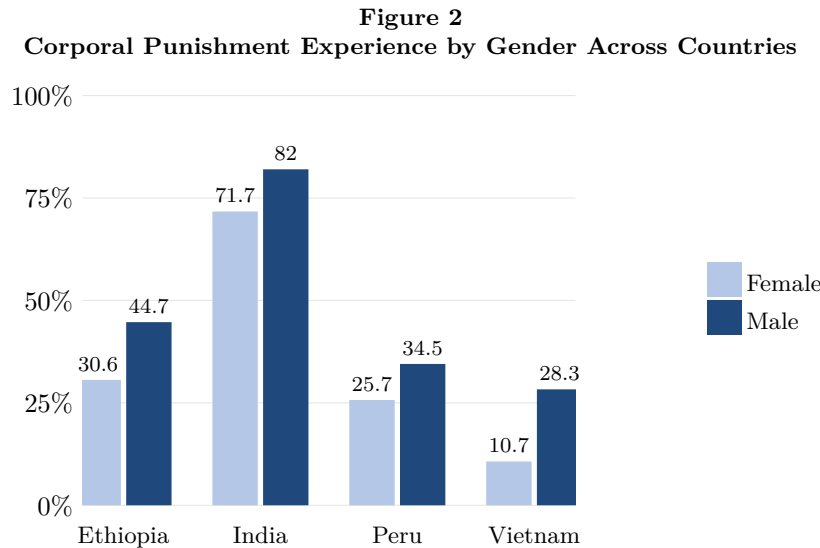
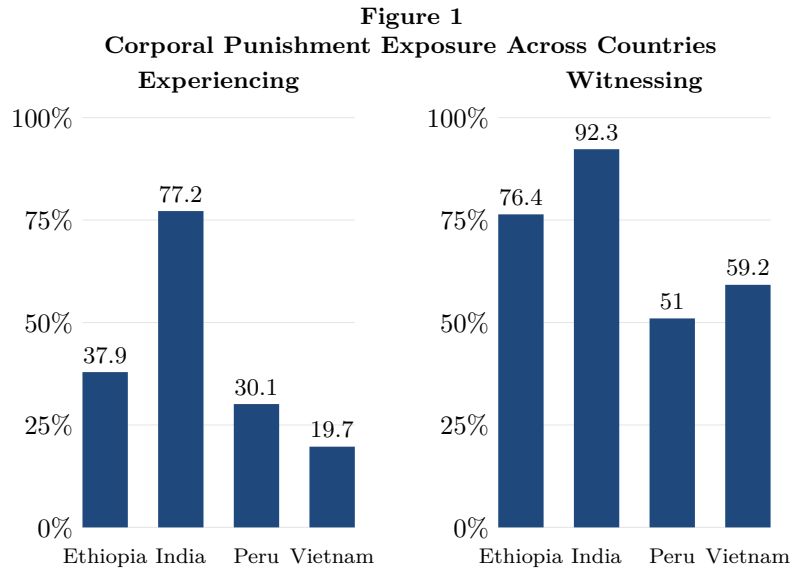


Figure 1 illustrate the differences in witnessing and experiencing corporal punishment, while Figure 2 looks at the rate by gender. There is a distinct cross-country pattern visible: boys tend to be hit at a higher rate than girls. The difference in gender is largest in Vietnam, with boys experiencing corporal punishment almost 3 times the rate of girls. This is a possible explanation as to why heterogeneous test results (5.2) shows a stronger effect for boys, as they are the more exposed group. Figure 3 examines corporal punishment rates by urban and rural location. Unlike gender, no consistent pattern emerges across countries: urban rates exceed rural rates in Ethiopia and Vietnam, while the reverse holds in India and Peru. This inconclusive pattern can possibly explain why the regression heterogeneity results for urban were weaker and more mixed than for gender. Rather than be able to draw a conclusion across all four countries, the urban/rural difference in rates is more dependent on a national context.

Figure 3
Rates of Corporal Punishment by Location

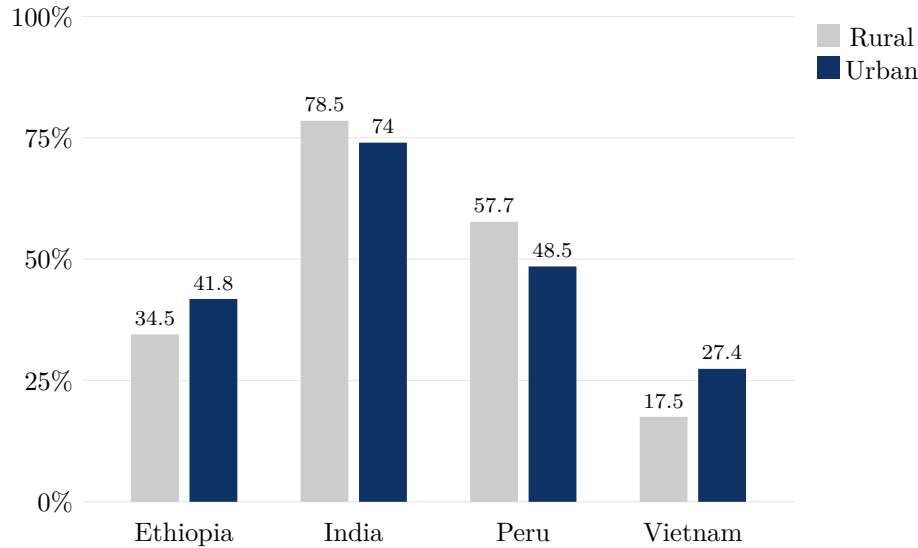
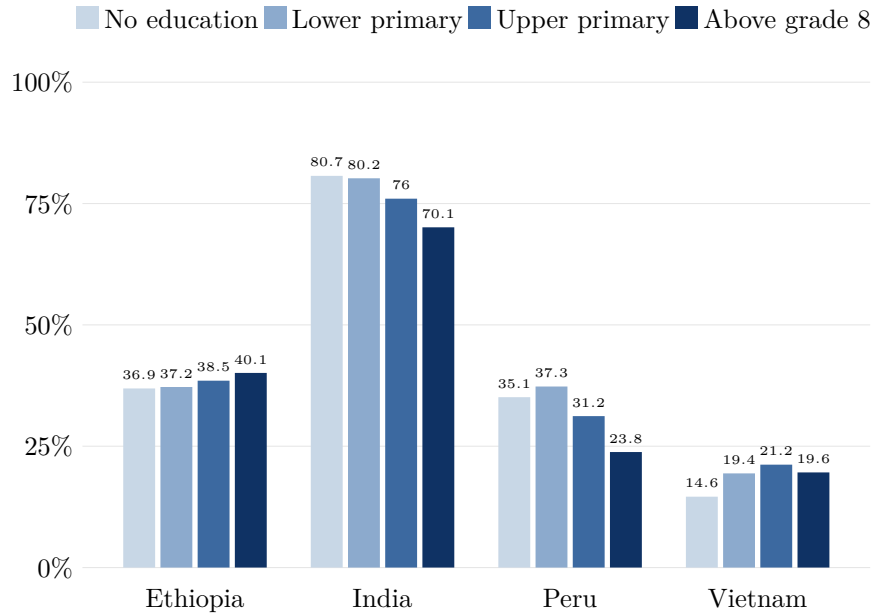


Figure 4 disaggregates the rates of CP by caregiver education levels. Similar to figure three, there's no cross country pattern that we can see. In Ethiopia, CP actually increases slightly with caregiver education; in India it decreases slightly; in Vietnam there's no clear pattern. The conclusion we can take away from this is that parents' level of education is not enough to stop CP, impact rates cut through all education levels.

Figure 4
Caregiver education level shows no consistent relationship with corporal punishment.



5.2 OLS and Cluster Fixed Effects Estimates

This section details the primary empirical results of the study, organized by outcome domain. We first present baseline OLS estimates alongside cluster fixed effects specifications for cognitive and psychosocial outcomes at age 15. The analysis then extends to early adulthood, examining the associations between early exposure to corporal punishment and developmental outcomes measured at ages 19-22.

5.2.1 Cognitive Outcomes

The OLS and fixed effects (FE) estimates, presented in Table 6, indicate a negative relationship between corporal punishment and cognitive performance at Round 5, with the strongest effects observed in Peru and Vietnam. In Peru, the OLS coefficient of -0.09 (SE = 0.04, $p < 0.05$) increases in magnitude to -0.11 (SE = 0.03, $p < 0.01$) when incorporating cluster fixed effects. The consistency of these estimates across both specifications suggests the association is not sensitive to the absorption of cluster-level confounders. Decomposing the composite into its math and vocabulary sub-components reinforces this picture: both scores are negative and significant across OLS and FE in Peru.

Table 6
Impact of Corporal Punishment on Cognitive Outcomes

	Peru		Vietnam		India		Ethiopia	
	OLS	FE	OLS	FE	OLS	FE	OLS	FE
Panel A. Cognitive Performance	-0.09** (0.04)	-0.11*** (0.03)	-0.08 (0.05)	-0.09* (0.04)	-0.02 (0.06)	-0.03 (0.06)	0.02 (0.04)	-0.02 (0.04)
N	1803	1803	1759	1759	1804	1804	1162	1162
R-Squared	0.51	0.49	0.32	0.40	0.34	0.39	0.55	0.59
A1. Math Score (Std.)	-0.08** (0.04)	-0.10** (0.04)	-0.16** (0.05)	-0.14** (0.05)	-0.03 (0.04)	-0.16*** (0.05)	-0.06 (0.05)	-0.07 (0.05)
N	1803	1803	1737	1737	1814	1822	1,245	1,245
R-Squared	0.36	0.33	0.31	0.35	0.36	0.13	0.42	0.44
A2. Vocabulary Score (Std.)	-0.07* (0.04)	-0.08** (0.03)	0.01 (0.06)	-0.03 (0.04)	-0.03 (0.06)	-0.11** (0.05)	0.08 (0.05)	0.02 (0.05)
N	1782	1782	1756	1756	1858	1864	1,191	1,191
R-Squared	0.48	0.47	0.16	0.30	0.13	0.09	0.44	0.51

In Vietnam, the composite OLS estimate for cognitive performance falls marginally short of conventional significance; however, the FE estimate crosses the threshold, suggesting that within-cluster variation more clearly identifies the negative impact of corporal punishment. Decomposing the composite reveals this is driven almost exclusively by math performance, where both OLS and FE estimates are large and significant at -0.16 and -0.14, respectively. This concentration of harm in numeracy rather than literacy distinguishes Vietnam context from Peru and may carry particular implications for future labour market outcomes, given that numeracy is among the strongest predictors of earnings in low- and middle-income contexts (Bhula et al., 2025).

India shows no significant association at the composite level under OLS, yet the FE estimate for math performance is large and significant (-0.16, SE = 0.049, $p < 0.001$). This divergence likely reflects upward bias in the OLS estimates, where communities with high corporal punishment prevalence may share characteristics that independently support math learning. Furthermore, the high prevalence of corporal punishment in the Indian sample (77 percent) may further limit identifying variation in OLS specifications. Finally, Ethiopia shows no significant association across any specification or sub-components.

5.2.2 Psychosocial Outcomes

Table 7 indicates that Vietnam exhibits the most consistent psychosocial effects. The estimates for composite psychosocial performance are -0.14 under OLS and -0.15 under fixed effects. Disaggregating by sub-index, the agency index and self-esteem index drive the composite result. The shame and self-efficacy indices are statistically insignificant across both specifications, suggesting that the psychosocial effects of corporal punishment in Vietnam operate primarily through reductions in agency and self-esteem rather than through shame or self-efficacy.

Estimates for Peru, India, and Ethiopia are statistically insignificant across all psychosocial outcomes and both specifications, with the exception of India's agency index.

Table 7
Impact of Corporal Punishment on Psychosocial Outcomes

	Peru		Vietnam		India		Ethiopia	
	OLS	FE	OLS	FE	OLS	FE	OLS	FE
Panel B. Psychosocial Performance	-0.02 (0.03)	-0.02 (0.02)	-0.14** (0.07)	-0.15** (0.06)	-0.01 (0.05)	-0.01 (0.05)	-0.09 (0.05)	-0.08 (0.05)
N	1792	1792	1758	1758	1804	1804	1279	1279
R-Squared	0.00	0.00	0.10	0.14	0.06	0.11	0.12	0.22
B1. Agency Index	-0.08 (0.05)	-0.07 (0.04)	-0.09 (0.06)	-0.11** (0.05)	-0.07*** (0.03)	-0.10** (0.04)	-0.08 (0.05)	-0.02 (0.05)
N	1790	1790	1758	1758	1861	1867	1,279	1,279
R-Squared	0.13	0.13	0.12	0.18	0.08	0.03	0.09	0.20
B2. Shame Index	0.05 (0.05)	0.06 (0.05)	0.09 (0.06)	0.10 (0.07)	-0.10 (0.07)	-0.03 (0.05)	0.05 (0.07)	0.09 (0.06)
N	1790	1790	1757	1757	1862	1867	1,279	1,279
R-Squared	0.03	0.03	0.04	0.08	0.04	0.11	0.11	0.20
B3. Self-Efficacy Index	0.04 (0.05)	0.03 (0.05)	-0.06 (0.06)	-0.07 (0.07)	-0.04 (0.08)	-0.07 (0.10)	-0.07 (0.07)	-0.06 (0.04)
N	1791	1791	1758	1758	1861	1866	1,279	1,279
R-Squared	0.07	0.07	0.03	0.08	0.02	0.04	0.06	0.16
B4. Self-Esteem Index	-0.63 (0.58)	-0.02 (0.02)	-0.11* (0.06)	-0.11* (0.06)	0.00 (0.06)	-0.01 (0.06)	-0.02 (0.06)	-0.01 (0.07)
N	1791	1792	1758	1758	1806	1867	1,279	1,279
R-Squared	0.00	0.00	0.03	0.06	0.00	0.05	0.05	0.16

5.2.3 Later Life Outcomes

Table 8 reports the associations between corporal punishment and outcomes measured in early adulthood (Round 6), connecting the developmental patterns documented at Round 5 to longer-run consequences.

In Vietnam, corporal punishment is associated with a 9 percentage point reduction in the educational enrollment rate under OLS and 7 percentage points under fixed effects, both significant at the 5 percent level. The Vietnamese result is consistent with the Round 5 psychosocial findings: children whose sense of agency and self-esteem has been eroded are less likely to persist in schooling.

In Peru, corporal punishment is associated with a 6 percentage point higher employment rate under both specifications, significant at the 1 percent level under OLS and the 5 percent level under fixed effects.

All remaining estimates are statistically insignificant at conventional levels, including anxiety and depression across all four countries, enrollment in Peru, India, and Ethiopia, and employment in Vietnam, India, and Ethiopia.

Table 8
Impact of Corporal Punishment on Later Life Outcomes

	Peru		Vietnam		India		Ethiopia	
	OLS	FE	OLS	FE	OLS	FE	OLS	FE
C1. Anxiety	0.13 (0.19)	0.12 (0.13)	0.10 (0.13)	0.01 (0.13)	0.02 (0.10)	0.21 (0.13)	0.14 (0.13)	0.15 (0.13)
N	1540	1540	1553	1553	1774	1774	1193	1193
R-Squared	0.07	0.07	0.03	0.16	0.09	0.23	0.14	0.39
C2. Depression	-0.07 (0.18)	-0.08 (0.13)	0.29* (0.14)	0.15 (0.11)	-0.10 (0.08)	0.12 (0.10)	-0.01 (0.14)	-0.02 (0.10)
N	1540	1540	1553	1553	1774	1774	1193	1193
R-Squared	0.05	0.05	0.02	0.19	0.10	0.28	0.19	0.47
C3. Education Enrollment Rate	-0.03 (0.03)	-0.04 (0.03)	-0.09** (0.04)	-0.07** (0.04)	0.01 (0.03)	-0.01 (0.03)	0.01 (0.02)	-0.01 (0.02)
N	1543	1543	1553	1553	1774	1774	1193	1193
R-Squared	0.07	0.07	0.27	0.32	0.15	0.21	0.22	0.37
C4. Employment Rate	0.06*** (0.02)	0.06** (0.02)	0.05 (0.03)	0.04 (0.03)	0.02 (0.03)	0.03 (0.03)	0.03 (0.02)	0.37 (0.02)
N	1543	1543	1553	1553	1774	1774	1193	1193
R-Squared	0.14	0.14	0.14	0.18	0.30	0.38	0.19	0.32

5.3 Instrumental Variables (IV) Estimates

The IV strategy uses leave-one-out measures of corporal punishment witnessed by other children in the same community as an instrument for a child's own exposure. A key requirement is a strong first stage. As shown in Tables A7 and A8, the first-stage F-statistics are above conventional thresholds in Peru ($F \approx 19$) and Vietnam ($F \approx 22$), suggesting that the instrument is sufficiently strong in these contexts.

In contrast, the first stage is weaker in India and Ethiopia, and remain weak for later-life outcomes across most countries with the exception of Peru (Table A9). IV estimates for countries and outcomes where the instrument is weak should therefore be interpreted with caution, and the OLS and fixed effects estimates remain the preferred specifications in these cases.

5.3.1 Cognitive and Psychosocial Outcomes

In Vietnam, the instrumental variables estimates reinforce the adverse impacts on numeracy previously discussed; the math score coefficient increases substantially in magnitude to -2.04 , compared to the baseline OLS and cluster fixed effects estimates (Table A7). For psychosocial outcomes, significant effects are concentrated in Peru, where corporal punishment results in a 1.26 standard deviation reduction in the Shame Index, significant at the 5 percent level (Table A8). Interestingly, these significant IV estimates are considerably larger in magnitude than their corresponding OLS and FE counterparts.

5.3.2 Later Life Outcomes

Table A9 reports IV estimates of the impact of corporal punishment on later life outcomes. For mental health, India's IV estimate for anxiety is negative and statistically significant; however, the first-stage F-statistic of 7.19 indicates a weak instrument, and the estimate is unlikely to support a causal interpretation. Vietnam's IV estimate for depression is positive and statistically significant, though the first-stage F-statistic of 9.86 again falls marginally below the conventional threshold.

For educational enrollment, the IV estimate for Vietnam is negative and marginally significant, consistent with the pattern observed in the OLS and fixed effects results. In contrast, the IV estimate for Peru is positive and statistically significant, but this is difficult to reconcile with the OLS and fixed effects estimates and likely reflects weak instrument

bias rather than a causal effect. For employment, there is no evidence of statistically significant effects in any country under the IV specification.

5.4 Heterogeneity Analysis

The heterogeneity analysis examines whether the impact of corporal punishment varies across three dimensions: gender, location, and caregiver education. Of the four countries, heterogeneity effects are concentrated in Peru, with limited evidence across Vietnam, India, and Ethiopia. For this reason, Peru's results are displayed below, while the results for Vietnam, Ethiopia and India can be found in the Appendix.

5.4.1 Cognitive and Psychosocial Performance

In Peru, the baseline effect for females is -0.16 , while the interaction term Corporal Punishment $\times$ Male is 0.11 (Table 9), indicating that the adverse cognitive association is significantly attenuated for male students. Location also moderates the association: the Corporal Punishment $\times$ Urban interaction is -0.17 , suggesting that urban children face a larger cognitive penalty from corporal punishment than their rural counterparts. For psychosocial performance, no significant heterogeneous effects are detected across any subgroup or country at the 5 percent significance level.

5.4.2 Later Life Outcomes

For later life outcomes in Peru, heterogeneous effects are concentrated in the gender subgroup. For anxiety (Table 11), the baseline effect for females is 0.79 , while the Corporal Punishment $\times$ Male interaction is -1.51 , indicating that the negative anxiety association is significantly larger for females than for males. A similar pattern holds for depression (Table 12): the baseline effect for females is 0.54 and the Corporal Punishment $\times$ Male interaction is -1.31 , confirming that the mental health consequences of corporal punishment are concentrated among female students.

For educational enrollment (Table 13), the Corporal Punishment $\times$ Male interaction is -0.13 , indicating that male students exposed to corporal punishment face a larger decline in enrollment relative to female students. For employment (Table 14), the interaction terms are positive for both males (0.20) and urban students (0.15), suggesting the positive association between corporal punishment and employment is stronger for these groups.

Table 9
Heterogeneous Effects on Cognitive Performance, Peru

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	-0.162*** (0.035)	0.000 (0.051)	-0.149*** (0.043)
Corporal Punishment × Male	0.107*** (0.037)		
Corporal Punishment × Urban		-0.167** (0.074)	
Caregiver Education: Corporal Punishment × Above Primary			0.065 (0.075)
Observations	1803	1803	1803
R-squared	0.51	0.52	0.52

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10
Heterogeneous Effects on Psychosocial Performance, Peru

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	-0.011 (0.020)	0.014 (0.016)	-0.001 (0.012)
Corporal Punishment × Male	0.002 (0.015)		
Corporal Punishment × Urban		-0.023 (0.028)	
Caregiver Education: Corporal Punishment × Above Primary			-0.002 (0.036)
Observations	1792	1792	1792
R-squared	0.01	0.01	0.01

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11
Heterogeneous Effects on Anxiety (GAD-7), Peru

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	0.794*** (0.251)	0.128 (0.273)	0.194 (0.166)
Corporal Punishment $\times$ Male	-1.510*** (0.340)		
Corporal Punishment $\times$ Urban		-0.035 (0.343)	
Caregiver Education: Corporal Punishment $\times$ Above Primary			0.203 (0.310)
Observations	1540	1540	1540
R-squared	0.06	0.10	0.09

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12
Heterogeneous Effects on Depression (PHQ-8), Peru

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	0.535** (0.205)	0.064 (0.256)	0.017 (0.179)
Corporal Punishment $\times$ Male	-1.314*** (0.278)		
Corporal Punishment $\times$ Urban		-0.207 (0.271)	
Caregiver Education: Corporal Punishment $\times$ Above Primary			-0.228 (0.273)
Observations	1540	1540	1540
R-squared	0.08	0.10	0.09

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13
Heterogeneous Effects on Education Enrollment, Peru

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	0.026 (0.037)	0.062 (0.047)	−0.011 (0.036)
Corporal Punishment × Male	−0.125*** (0.041)		
Corporal Punishment × Urban		−0.133* (0.371)	
Caregiver Education: Corporal Punishment × Above Primary			−0.011 (0.036)
Observations	1543	1543	1543
R-squared	0.12	0.13	0.12

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14
Heterogeneous Effects on Employment, Peru

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	−0.027 (0.034)	−0.047 (0.055)	0.064** (0.030)
Corporal Punishment × Male	0.196*** (0.042)		
Corporal Punishment × Urban		0.153** (0.059)	
Caregiver Education: Corporal Punishment × Above Primary			0.001 (0.038)
Observations	1543	1543	1543
R-squared	0.13	0.17	0.16

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

6. Limitations and Discussion

While the results provide useful evidence, several limitations should be noted. This section discusses these limitations and places the findings in context.

6.1 Limitations and Directions for Future Research

This study is subject to three primary limitations. First, despite our identification strategy, the possibility of omitted variable bias cannot be entirely dismissed. Due to the constraints of the available data, the initial set of covariates provided to the PDS LASSO is restricted to child and household characteristics. Future research would benefit from incorporating school-level characteristics and other inputs from the educational production function (Glewwe and Kremer, 2006).

Second, corporal punishment was originally coded as an ordinal variable (0 = never; 1 = once or twice; 2 = most or all of the time) and recoded as a binary indicator. Future research could explore the ordinal variation. Additionally, this study focuses on corporal punishment administered at school, whereas the psychology and public health literature examines punishment at home. Incorporating domestic corporal punishment would provide a more complete picture of its cumulative effect on child developmental outcomes.

Third, the instrumental variable is constructed as the average corporal punishment witnessed by other children in the same community, which yields a weak instrument. The community level could be too broad to capture the propensity for corporal punishment experienced by a given child, as the relevant variation may operate at the classroom level. Future research could explore more granular identifiers or alternative instruments to improve the credibility of IV estimates.

6.2 Persistence and Attenuation Over Time

The results are consistent with the existing literature on the adverse effects of school-based corporal punishment. Negative associations with cognitive performance are observed in Peru, and with psychosocial performance in Vietnam, while math scores show adverse associations across most countries with the exception of Ethiopia. The persistence of these associations from age 8 to age 15, a seven-year window, suggests that early developmental harm is durable rather than transitory.

By early adulthood (Round 6, ages 19-22), most coefficients attenuate toward zero and lose statistical significance, with the exceptions of educational enrollment in Vietnam and employment in Peru. At least two mechanisms may account for this attenuation.

First, attrition in Round 6 appears to be non-random. In Vietnam, for instance, children who attrited scored lower in math and exhibited a lower agency index at baseline relative to those who remained in the sample. If the most adversely affected children are disproportionately absent from Round 6, the estimated associations will understate the true long-run effects, even after reweighting to recover baseline representativeness. This suggests that the null later-life results for most countries in Round 6 should be interpreted with caution as potential underestimates of the true impact, rather than definitive evidence of developmental recovery. Second, the interval between the initial disciplinary shock and the measurement of adult outcomes allows for confounding life events, such as labor market entry, family formation, and shifts in social networks, to dilute the early childhood shock. The positive association with employment in Peru and the negative association with school enrollment in Vietnam are consistent with this interpretation. Rather than reflecting recovery, these patterns suggest a reallocation of human capital investment from schooling to early labour market participation among those exposed to corporal punishment.

6.3 Cross-Country Variation and Contextual Factors

The results for both rounds vary substantially across countries, highlighting the importance of contextual and institutional factors. Despite a relatively lower prevalence of around 20 percent of the sample and the formal legal prohibition of corporal punishment in schools, Vietnam shows the strongest and most consistent negative effects. This could mean that students who experienced violence in schools faced more intense disruptions due to higher marginal impact. In contrast, Ethiopia and India show weaker and less consistent results, diverging from earlier literature. This could be as a result of the normalization of corporal punishment, where the widespread use of violence in schools reduces variation and makes it more difficult to detect effects empirically.

7. Conclusion and Policy Implications

This study provides evidence that exposure to corporal punishment in childhood is associated with weaker cognitive and psychosocial outcomes in adolescence, with more limited effects on early-adulthood outcomes. While the magnitude and consistency of effects vary across countries and specifications, the results suggest that early exposure to violence in schools may have lasting implications for human capital formation.

These findings have several policy implications. They highlight that legislation alone is unlikely to be sufficient. Despite the formal legal prohibition of corporal punishment in many countries, the practice persists in classrooms. Governments should move beyond legislation alone and invest in implementation.

At the school level, corporal punishment is often associated with structural constraints such as large class sizes and limited teacher training. Policies that strengthen teacher capacity, particularly in classroom management and positive discipline techniques, may reduce reliance on physical punishment while improving the learning environment.

At the household and community levels, corporal punishment is often sustained by social norms that view it as necessary for discipline. Public awareness campaigns and community engagement can reshape these norms by emphasizing the long-term harms and promoting non-violent alternatives.

Ultimately, reducing corporal punishment requires a combination of policy tools that integrates effective legal frameworks, meaningful investments in teacher support, and thoughtful efforts to shift social norms toward more positive and constructive forms of discipline.

References

- Anderson, M. L. (2008). [Multiple inference and gender differences in the effects of early intervention: A reevaluation of the Abecedarian, Perry Preschool, and Early Training projects](#), *Journal of the American Statistical Association*, 103(484), 1481–1495.
- Baker-Henningham, H., Meeks-Gardner, J., Chang, S. and Walker, S. (2009). [Experiences of violence and deficits in academic achievement among urban primary school children in jamaica](#), *Child Abuse & Neglect*, 33(5), 296–306.
- Bangash, M., Nayab, T., Zubair, M., Nawaz, H., Ullah, I. and Ullah, S. (2024). [Examining the impacts of corporal punishment in secondary schools: Unraveling psychological repercussions on students](#), *Journal of Health and Rehabilitation Research*, 4(2), 364–369.
- Belloni, A., Chernozhukov, V. and Hansen, C. (2014). [Inference on treatment effects after selection among high-dimensional controls](#), *Review of Economic Studies*, 81, 608–650.
- Bhula, R., Glennerster, R. and Puri, J. (2025). The economic returns to foundational literacy and numeracy: Evidence from indonesia, *Technical report*, Center for Global Development.
- Durrant, J. E. and Ensom, R. (2020). [An Eriksonian perspective on physical punishment and its impacts on mental health](#), *The Psychoanalytic Study of the Child*, 73(1), 30–45.
- Elendu, C., Amaechi, D. C., Elendu, T. C., Olabode, D. M., Elendu, I. D., Amaechi, E. C., Aggreh, O. P. and Bright, C. C. (2025). [Ending the practice of corporal punishment in Nigeria: A call for progress and dignity](#), *Annals of Medicine & Surgery*, 87, 5688–5695.
- Evans, D. and Hares, S. (2023). School violence and learning outcomes: Beyond WDR 2018, World Bank Blogs: Development Talk.
- Federal Democratic Republic of Ethiopia (2005). Criminal code of the Federal Democratic Republic of Ethiopia.
- Ferrara, P., Franceschini, G., Villani, A. et al. (2019). [Physical, psychological, and social impact of school violence on children](#), *Italian Journal of Pediatrics*, 45, 76.

- Gershoff, E. T. (2002). [Corporal punishment by parents and associated child behaviors and experiences: A meta-analytic and theoretical review](#), *Psychological Bulletin*, 128(4), 539–579.
- Glewwe, P. and Kremer, M. (2006). [Schools, teachers, and education outcomes in developing countries](#), in E. A. Hanushek and F. Welch (eds), *Handbook of the Economics of Education*, Elsevier, Amsterdam, pp. 945–1017.
- Global Initiative to End All Corporal Punishment of Children (2010). Briefing on Ethiopia for the Human Rights Council, Briefing Paper.
- Global Initiative to End All Corporal Punishment of Children (2014a). Briefing on Ethiopia for the committee on the rights of the child / upr-related briefing.
- Global Initiative to End All Corporal Punishment of Children (2014b). Corporal punishment of children in Viet Nam. Country report (accessed 2026-04-17).
- Global Initiative to End All Corporal Punishment of Children (2020). Global report 2019: Progress towards ending corporal punishment of children, *Technical report*, Global Initiative to End All Corporal Punishment of Children.
- Government of India (2009). The right of children to free and compulsory education act.
- Hecker, T., Hermenau, K., Isele, D. and Elbert, T. (2014). [Corporal punishment and children's externalizing problems: A cross-sectional study of Tanzanian primary school aged children](#), *Child Abuse & Neglect*, 38(5), 884–892.
- Heckman, J. J. (2011). The economics of inequality: The value of early childhood education, *American Educator*, 35(1), 31–35.
- Lakshman, I. M. (2018). [Can Sri Lankan teachers afford to spare the rod? teacher attitudes towards corporal punishment in school](#), *Cogent Social Sciences*, 4(1), 1536316.
- Lokot, M., Bhatia, A., Kenny, L. and Cislighi, B. (2020). [Corporal punishment, discipline and social norms: A systematic review in low- and middle-income countries](#), *Aggression and Violent Behavior*, 55, 101507.

- Maiti, A. (2021). [Effect of corporal punishment on young children's educational outcomes](#), *Education Economics*, 29(4), 411–423.
- Merrill, K. G., Knight, L., Glynn, J. R., Allen, E., Naker, D. and Devries, K. M. (2017). [School staff perpetration of physical violence against students in Uganda: A multilevel analysis of risk factors](#), *BMJ Open*, 7(8), e015567.
- Ministry of Education and Training of Vietnam (2008). Decision no. 16/2008/qd-bgddt promulgating the regulation on the teaching professionals' moral standards. Accessed 2026-04-17.
- Naz, A., Khan, W., Daraz, U., Hussain, M. and Khan, Q. (2011). The impacts of corporal punishment on students' academic performance/-career and personality development up-to secondary level education in Khyber Pakhtunkhwa pakistan, *International Journal of Business and Social Science*, 2(12), 130–139.
- Oganda Portela, M. J. and Pells, K. (2015). Corporal punishment in schools: Longitudinal evidence from Ethiopia, India, Peru and Viet Nam, *Innocenti Discussion Paper 2015-02*, UNICEF Office of Research.
- Paolucci, E. O. and Violato, C. (2004). [A meta-analysis of the published research on the affective, cognitive, and behavioral effects of corporal punishment](#), *The Journal of Psychology: Interdisciplinary and Applied*, 138(3), 197–222.
- Pinheiro, P. S. (2006). *World Report on Violence against Children*, United Nations, Geneva.
- Psaki, S. R., Mensch, B. S. and Soler-Hampejsek, E. (2017). [Associations between violence in school and at home and education outcomes in rural Malawi: A longitudinal analysis](#), *Comparative Education Review*, 61(2), 354–390.
- Schulte, M.-C. (2021). [Corporal punishment in schools in low-income countries: An evidence review](#), *Technical report*, USAID / NORC at the University of Chicago.
- Smith, A. B. (2006). The state of research on the effects of physical punishment, *Social Policy Journal of New Zealand*, (27), 114–127.

- Ssenyonga, J., Hermenau, K., Nkuba, M. and Hecker, T. (2019). [Stress and positive attitudes towards violent discipline are associated with school violence by Ugandan teachers](#), *Child Abuse & Neglect*, 93, 15–26.
- UN Committee on the Rights of the Child (2006). General comment no. 8 (2006): The right of the child to protection from corporal punishment and other cruel or degrading forms of punishment (arts. 19; 28, para. 2; and 37, inter alia), *Technical Report CRC/C/GC/8*, United Nations.
- World Health Organization (2025). Corporal punishment of children: The public health impact, *Technical Report B09424*, World Health Organization.
- Young Lives (2015). Corporal punishment of children linked to lower school grades. Accessed 2026-04-17.
- Zhang, H., Wang, Z., Tang, X. and Wang, W. (2025). [The association between child maltreatment and academic achievement: A systematic review and meta-analysis](#), *Child Abuse & Neglect*, 159, 107159.
- Zuberi, T., Seni, A. J. and Sungwa, R. (2025). [Understanding perceptions of teachers and pupils towards corporal punishment for pupils with disabilities in primary schools in Tanzania: Implications for the promotion of social justice in education](#), *International Journal of Educational Innovation and Research*, 4(1), 58–73.

A. Appendix

Table A1
Attrition Analysis, Ethiopia

Variable	Group 1 (Not Attrite)		Group 2 (Attrite)		(1)-(2)	
	N	Mean (SE)	N	Mean (SE)	N	Mean Diff.
<i>Panel A. Round 5</i>						
Math Score (Std.)	1395	-0.010 (0.026)	35	0.395 (0.233)	1430	-0.405**
Vocabulary Score (Std.)	1388	-0.005 (0.027)	35	0.190 (0.191)	1423	-0.195
Agency Index	1410	0.157 (0.025)	35	0.230 (0.156)	1445	-0.072
Shame Index	1410	-0.074 (0.025)	35	0.289 (0.186)	1445	-0.363**
Male	1410	0.519 (0.013)	35	0.457 (0.085)	1445	0.062
Corporal Punishment	1410	0.379 (0.013)	35	0.343 (0.081)	1445	0.037
<i>Panel B. Round 6</i>						
Math Score (Std.)	1283	-0.002 (0.028)	147	0.015 (0.091)	1430	-0.017
Vocabulary Score (Std.)	1277	0.003 (0.028)	146	-0.022 (0.082)	1423	0.025
Agency Index	1295	0.148 (0.026)	150	0.250 (0.074)	1445	-0.102
Shame Index	1295	-0.049 (0.027)	150	-0.200 (0.073)	1445	0.150*
Male	1295	0.520 (0.014)	150	0.500 (0.041)	1445	0.020
Corporal Punishment	1295	0.380 (0.013)	150	0.367 (0.039)	1445	0.013

* p<0.1, ** p<0.05, *** p<0.01.

Table A2
Attrition Analysis, India

Variable	Group 1 (Not Attrite)		Group 2 (Attrite)		(1)-(2)	
	N	Mean (SE)	N	Mean (SE)	N	Mean Diff.
<i>Panel A. Round 5</i>						
Math Score (Std.)	1865	0.004 (0.023)	29	-0.236 (0.205)	1894	0.240
Vocabulary Score (Std.)	1857	-0.002 (0.023)	27	0.142 (0.227)	1884	-0.144
Agency Index	1869	0.006 (0.023)	29	-0.363 (0.185)	1898	0.368**
Shame Index	1870	-0.001 (0.023)	29	0.044 (0.172)	1899	-0.045
Male	1872	0.538 (0.012)	29	0.414 (0.093)	1901	0.125
Corporal Punishment	1872	0.774 (0.010)	29	0.690 (0.087)	1901	0.084
<i>Panel B. Round 6</i>						
Math Score (Std.)	1834	0.003 (0.023)	60	-0.104 (0.119)	1894	0.107
Vocabulary Score (Std.)	1825	-0.005 (0.023)	59	0.156 (0.132)	1884	-0.161
Agency Index	1838	0.000 (0.023)	60	0.012 (0.128)	1898	-0.012
Shame Index	1839	-0.002 (0.023)	60	0.055 (0.118)	1899	-0.056
Male	1841	0.536 (0.012)	60	0.550 (0.065)	1901	-0.014
Corporal Punishment	1841	0.776 (0.010)	60	0.650 (0.062)	1901	0.126**

* p<0.1, ** p<0.05, *** p<0.01.

Table A3
Attrition Analysis, Peru

Variable	Group 1 (Not Attrite)		Group 2 (Attrite)		(1)-(2)	
	N	Mean (SE)	N	Mean (SE)	N	Mean Diff.
<i>Panel A. Round 5</i>						
Math Score (Std.)	1770	0.492 (0.005)	93	0.489 (0.021)	1863	0.003
Vocabulary Score (Std.)	1724	0.469 (0.003)	93	0.470 (0.015)	1817	0.001
Agency Index	1802	0.579 (0.003)	95	0.585 (0.011)	1897	0.006
Shame Index	1802	0.288 (0.003)	95	0.293 (0.014)	1897	0.004
Male	1803	0.503 (0.012)	95	0.505 (0.052)	1898	0.002
Corporal Punishment	1803	0.167 (0.006)	95	0.189 (0.031)	1898	-0.023
<i>Panel B. Round 6</i>						
Math Score (Std.)	1692	0.499 (0.005)	171	0.423 (0.015)	1863	0.077***
Vocabulary Score (Std.)	1650	0.472 (0.003)	167	0.432 (0.012)	1817	0.040***
Agency Index	1724	0.582 (0.003)	173	0.562 (0.008)	1897	0.020**
Shame Index	1724	0.287 (0.003)	173	0.300 (0.010)	1897	-0.013
Male	1725	0.497 (0.012)	173	0.561 (0.038)	1898	-0.063
Corporal Punishment	1725	0.166 (0.006)	173	0.182 (0.022)	1898	-0.016

* p<0.1, ** p<0.05, *** p<0.01.

Table A4
Attrition Analysis, Vietnam

Variable	Group 1 (Not Attrite)		Group 2 (Attrite)		(1)-(2)	
	N	Mean (SE)	N	Mean (SE)	N	Mean Diff.
<i>Panel A. Round 5</i>						
Math Score (Std.)	1838	-0.004 (0.023)	33	0.228 (0.198)	1871	-0.232
Vocabulary Score (Std.)	1748	-0.004 (0.024)	32	0.196 (0.227)	1780	-0.200
Agency Index	1839	0.001 (0.023)	34	-0.042 (0.163)	1873	0.043
Shame Index	1845	0.001 (0.023)	34	-0.039 (0.178)	1879	0.040
Male	1848	0.510 (0.012)	34	0.500 (0.087)	1882	0.010
Corporal Punishment	1848	0.194 (0.009)	34	0.324 (0.081)	1882	-0.129*
<i>Panel B. Round 6</i>						
Math Score (Std.)	1659	0.018 (0.024)	212	-0.143 (0.084)	1871	0.161**
Vocabulary Score (Std.)	1581	0.009 (0.025)	199	-0.070 (0.081)	1780	0.079
Agency Index	1658	0.021 (0.024)	215	-0.165 (0.070)	1873	0.187**
Shame Index	1664	0.004 (0.024)	215	-0.034 (0.073)	1879	0.038
Male	1667	0.498 (0.012)	215	0.600 (0.033)	1882	-0.102***
Corporal Punishment	1667	0.191 (0.010)	215	0.242 (0.029)	1882	-0.051*

* p<0.1, ** p<0.05, *** p<0.01.

Table A5
Corporal Punishment Prevalence
by Gender, Urban/Rural Status, and Caregiver Education

Subgroup	Country	Never (%)	Any CP (%)	N
<i>Panel A: By Gender</i>				
<i>Male</i>	Peru	65.55	34.45	943
	Vietnam	71.74	28.26	959
	India	18.04	81.96	1,020
	Ethiopia	55.30	44.70	748
<i>Female</i>	Peru	74.34	25.66	955
	Vietnam	89.27	10.73	923
	India	28.26	71.74	881
	Ethiopia	69.40	30.60	697
<i>Panel B: By Urban/Rural Status</i>				
<i>Urban</i>	Peru	51.55	48.45	1,352
	Vietnam	72.62	27.38	409
	India	25.99	74.01	481
	Ethiopia	58.20	41.80	663
<i>Rural</i>	Peru	42.35	57.65	451
	Vietnam	82.48	17.52	1,473
	India	21.53	78.47	1,403
	Ethiopia	65.50	34.50	782
<i>Panel C: By Caregiver Education</i>				
<i>No Education</i>	Peru	64.85	35.15	165
	Vietnam	85.42	14.58	192
	India	19.26	80.74	971
	Ethiopia	63.15	36.85	673
<i>Primary</i>	Peru	62.72	37.28	684
	Vietnam	80.56	19.44	504
	India	19.85	80.15	136
	Ethiopia	62.80	37.20	207
<i>Junior High / Upper Primary</i>	Peru	68.75	31.25	256
	Vietnam	78.79	21.21	844
	India	24.01	75.99	329
	Ethiopia	61.45	38.55	249
<i>Secondary Plus</i>	Peru	76.20	23.80	668
	Vietnam	80.43	19.57	327
	India	29.87	70.13	462
	Ethiopia	59.94	40.06	312

Note: Education categories harmonised across countries. Primary = grades 1–5 (Peru, Vietnam), grades 1–4 (India, Ethiopia). Secondary Plus = ≥10 years (Peru, Vietnam), above grade 8 (India, Ethiopia).

Table A6
Psychosocial Outcomes

Indicator	Question
Agency index	If I try hard, I can improve my situation in life I like to make plans for my future studies and work Other people in my family make all the decision about how I spend my time [recoded to positive] If I study hard at school, I will be rewarded by a better job in the future I have no choice about the work I do, I must do this sort of work [recoded to positive]
Shame index	I am proud of my clothes [recoded to negative] I am proud of my shoes or of having shoes [recoded to negative] I feel my clothing is right for all occasions [recoded to negative] I am never embarrassed because I do not have the right books, pencils or other equipment [recoded to negative] I am proud that I have the correct uniform [recoded to negative] I am proud of the work I have to do [recoded to negative]
General self-efficacy	I can always manage to solve difficult problems if I try hard enough If someone opposes me, I can find the means and ways to get what I want It is easy for me to stick to my aims and accomplish my goals I am confident that I could deal efficiently with unexpected events Thanks to my resourcefulness, I know how to handle unforeseen situations I can solve most problems if I invest the necessary effort I can remain calm when facing difficulties because I can rely on my coping abilities
General self-esteem	I do lots of important things In general, I like being the way I am Overall, I have a lot to be proud of I can do things as well as most people Other people think I am a good person A lot of things about me are good I'm as good as most other people When I do something, I do it well

Table A7
Impact of Corporal Punishment on Cognitive Outcomes (Round 5)

	Peru			Vietnam			India			Ethiopia		
	OLS	FE	IV	OLS	FE	IV	OLS	FE	IV	OLS	FE	IV
Panel A. Cognitive Performance												
Corporal punishment	-0.09** (0.04)	-0.11*** (0.03)	0.55 (0.41)	-0.08 (0.05)	-0.09* (0.04)	-1.41 (1.93)	-0.02 (0.06)	-0.03 (0.06)	2.01 (1.46)	0.02 (0.04)	-0.02 (0.04)	-0.57 (0.35)
N	1803	1803	1803	1759	1759	1663	1804	1804	1696	1162	1162	948
R-Squared	0.51	0.49	0.42	0.32	0.40		0.34	0.39		0.55	0.59	0.50
F-stat			19.85			21.64			6.59			2.58
A1. Math Score (Std.)												
Corporal punishment	-0.08** (0.04)	-0.10** (0.04)	0.75 (0.46)	-0.16** (0.05)	-0.14** (0.05)	-2.04* (1.19)	-0.03 (0.04)	-0.16*** (0.05)	-0.57 (1.13)	-0.06 (0.05)	-0.07 (0.05)	-0.50 (0.35)
N	1803	1803	1803	1737	1737	1642	1814	1822	1658	1245	1245	1024
R-Squared	0.36	0.33	0.22	0.31	0.35		0.36	0.13		0.42	0.44	0.41
F-stat			19.85			22.88			5.62			2.59
A2. Vocabulary Score (Std.)												
Corporal punishment	-0.07* (0.04)	-0.08** (0.03)	0.12 (0.42)	0.01 (0.06)	-0.03 (0.04)	-0.34 (2.29)	-0.03 (0.06)	-0.11** (0.05)	3.66 (1.95)	0.08 (0.05)	0.02 (0.05)	-0.62 (0.67)
N	1782	1782	1756	1756	1756	1661	1858	1864	1694	1191	1191	974
R-Squared	0.48	0.47	0.47	0.16	0.30		0.13	0.09		0.44	0.51	0.41
F-stat			18.40			22.07			6.54			3.27

Table A8
Impact of Corporal Punishment on Psychosocial Outcomes (Round 5)

	Peru			Vietnam			India			Ethiopia		
	OLS	FE	IV	OLS	FE	IV	OLS	FE	IV	OLS	FE	IV
Panel B. Psychosocial Performance												
Corporal punishment	-0.02 (0.03)	-0.02 (0.02)	-1.51 (1.45)	-0.14** (0.07)	-0.15** (0.06)	-1.46 (1.33)	-0.01 (0.05)	-0.01 (0.05)	2.32 (2.04)	-0.09 (0.05)	-0.08 (0.05)	-0.14 (0.42)
N	1792	1792	1792	1758	1758	1663	1804	1804	1697	1279	1279	1054
R-Squared	0.00	0.00		0.10	0.14		0.06	0.11		0.12	0.22	
F-stat			19.47			21.64			6.71			3.26
B1. Agency Index												
Corporal punishment	-0.08 (0.05)	-0.07 (0.04)	-1.06* (0.58)	-0.09 (0.06)	-0.11** (0.05)	-0.81 (1.22)	-0.07*** (0.03)	-0.10** (0.04)	-1.66 (0.94)	-0.08 (0.05)	-0.02 (0.05)	0.10 (0.42)
N	1790	1790	1790	1758	1758	1663	1861	1867	1697	1279	1279	1054
R-Squared	0.13	0.13		0.12	0.18		0.08	0.03		0.09	0.20	
F-stat			19.13			21.64			6.71			3.26
B2. Shame Index												
Corporal punishment	0.05 (0.05)	0.06 (0.05)	-1.26** (0.61)	0.09 (0.06)	0.10 (0.07)	0.27 (1.35)	-0.10 (0.07)	-0.03 (0.05)	-4.24 (2.87)	0.05 (0.07)	0.09 (0.06)	0.24 (0.39)
N	1790	1790	1790	1757	1757	1662	1862	1867	1697	1279	1279	1054
R-Squared	0.03	0.03		0.04	0.08		0.04	0.11		0.11	0.20	
F-stat			19.13			21.55			6.71			3.26
B3. Self-Efficacy Index												
Corporal punishment	0.04 (0.05)	0.03 (0.05)	-0.63 (0.58)	-0.06 (0.06)	-0.07 (0.05)	-1.69 (1.12)	-0.04 (0.08)	-0.07 (0.10)	1.56 (1.45)	-0.07 (0.04)	-0.06 (0.04)	-0.35 (0.41)
N	1791	1791	1791	1758	1758	1663	1861	1866	1696	1279	1279	1054
R-Squared	0.07	0.07		0.03	0.08		0.02	0.04		0.06	0.16	
F-stat			19.20			21.64			6.70			3.26
B4. Self-Esteem Index												
Corporal punishment	-0.63 (0.58)	-0.02 (0.02)	-1.51 (1.45)	-0.11* (0.06)	-0.11* (0.06)	-1.49* (0.87)	0.00 (0.06)	-0.01 (0.06)	2.55 (1.99)	-0.02 (0.06)	-0.01 (0.07)	0.17 (0.41)
N	1791	1792	1792	1758	1758	1663	1806	1867	1697	1279	1279	1054
R-Squared	0.00	0.00		0.03	0.06		0.00	0.05		0.05	0.16	
F-stat			19.47			21.64			6.71			3.26

Table A9
Impact of Corporal Punishment on Later-life Outcomes (Round 6)

Panel C. Later Life Outcomes	Peru			Vietnam			India			Ethiopia		
	OLS	FE	IV	OLS	FE	IV	OLS	FE	IV	OLS	FE	IV
C1. Anxiety												
Corporal Punishment	0.13 (0.19)	0.12 (0.13)	-3.35 (2.27)	0.10 (0.13)	0.01 (0.13)	4.64 (4.59)	0.02 (0.10)	0.21 (0.13)	-6.06** (2.53)	0.14 (0.13)	0.15 (0.13)	-8.49 (3.17)
N	1540	1540	1540	1553	1553	1468	1774	1774	1672	1193	1193	999
R-Squared	0.07	0.07		0.03	0.16		0.09	0.23		0.14	0.39	
F-stat			14.91			9.86			7.19			9.34
C2. Depression												
Corporal Punishment	-0.07 (0.18)	-0.08 (0.13)	-2.77 (2.11)	0.29* (0.14)	0.15 (0.11)	9.46** (4.46)	-0.10 (0.08)	0.12 (0.10)	-4.83 (2.84)	-0.01 (0.14)	-0.02 (0.10)	-4.13 (1.92)
N	1540	1540	1540	1553	1553	1468	1774	1774	1672	1193	1193	999
R-Squared	0.05	0.05		0.02	0.19		0.10	0.28		0.19	0.47	
F-stat			14.91			9.86			7.19			9.34
C3. Education Enrollment												
Corporal Punishment	-0.03 (0.03)	-0.04 (0.03)	0.91** (0.37)	-0.09** (0.04)	-0.07** (0.04)	-1.88* (0.97)	0.01 (0.03)	-0.01 (0.03)	0.89 (0.59)	0.01 (0.02)	-0.01 (0.02)	1.54 (0.57)
N	1543	1543	1543	1553	1553	1468	1774	1774	1672	1193	1193	999
R-Squared	0.07	0.07		0.27	0.32		0.15	0.21		0.22	0.37	
F-stat			14.87			9.86			7.19			9.34
C4. Employment												
Corporal Punishment	0.06*** (0.02)	0.06** (0.02)	-0.16 (0.25)	0.05 (0.03)	0.04 (0.03)	1.17* (0.65)	0.02 (0.03)	0.03 (0.03)	-0.54 (0.44)	0.03 (0.02)	-0.02 (0.02)	0.22 (0.30)
N	1543	1543	1543	1553	1553	1468	1774	1774	1672	1193	1193	999
R-Squared	0.14	0.14	0.09	0.14	0.18		0.30	0.38		0.19	0.32	
F-stat			14.87			9.86			7.19			9.34

Table A10
Heterogeneous Effects on Cognitive Performance, Ethiopia

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	−0.008 (0.058)	0.014 (0.055)	−0.025 (0.061)
Corporal Punishment × Male	−0.022 (0.060)		
Corporal Punishment × Urban		−0.067 (0.081)	
Caregiver Education: Corporal Punishment × Any Education			0.009 (0.076)
Observations	1162	1162	1162
R-squared	0.59	0.59	0.59

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A11
Heterogeneous Effects on Psychosocial Performance, Ethiopia

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	−0.106 (0.100)	−0.007 (0.081)	−0.156* (0.076)
Corporal Punishment × Male	0.052 (0.130)		
Corporal Punishment × Urban		−0.148 (0.113)	
Caregiver Education: Corporal Punishment × Any Education			0.150 (0.127)
Observations	1279	1279	1279
R-squared	0.22	0.23	0.23

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A12
Heterogeneous Effects on Anxiety (GAD-7), Ethiopia

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	0.381** (0.158)	0.159 (0.109)	0.166 (0.114)
Corporal Punishment $\times$ Male	-0.422* (0.226)		
Corporal Punishment $\times$ Urban		-0.030 (0.270)	
Caregiver Education: Corporal Punishment $\times$ Any Education			-0.040 (0.330)
Observations	1193	1193	1193
R-squared	0.39	0.39	0.39

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A13
Heterogeneous Effects on Depression (PHQ-8), Ethiopia

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	0.139 (0.114)	-0.035 (0.116)	-0.067 (0.139)
Corporal Punishment $\times$ Male	-0.281 (0.198)		
Corporal Punishment $\times$ Urban		0.034 (0.173)	
Caregiver Education: Corporal Punishment $\times$ Any Education			0.091 (0.275)
Observations	1193	1193	1193
R-squared	0.47	0.47	0.47

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A14
Heterogeneous Effects on Education Enrollment, Ethiopia

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	−0.002 (0.039)	0.015 (0.034)	0.021 (0.037)
Corporal Punishment × Male	−0.005 (0.046)		
Corporal Punishment × Urban		−0.043 (0.041)	
Caregiver Education: Corporal Punishment × Any Education			−0.049 (0.044)
Observations	1193	1193	1193
R-squared	0.37	0.38	0.38

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A15
Heterogeneous Effects on Employment, Ethiopia

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	−0.027 (0.044)	−0.040 (0.033)	−0.067** (0.032)
Corporal Punishment × Male	0.013 (0.058)		
Corporal Punishment × Urban		0.042 (0.046)	
Caregiver Education: Corporal Punishment × Any Education			0.088* (0.044)
Observations	1193	1193	1193
R-squared	0.32	0.32	0.33

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A16
Heterogeneous Effects on Cognitive Performance, India

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	−0.048 (0.065)	−0.048 (0.070)	−0.099 (0.087)
Corporal Punishment × Male	0.044 (0.086)		
Corporal Punishment × Urban		0.070 (0.103)	
Caregiver Education: Corporal Punishment × Any Education			0.131 (0.095)
Observations	1804	1804	1804
R-squared	0.39	0.39	0.39

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A17
Heterogeneous Effects on Psychosocial Performance, India

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	0.022 (0.082)	−0.044 (0.047)	−0.102 (0.064)
Corporal Punishment × Male	−0.072 (0.110)		
Corporal Punishment × Urban		0.118 (0.116)	
Caregiver Education: Corporal Punishment × Any Education			0.167 (0.102)
Observations	1804	1804	1804
R-squared	0.11	0.11	0.12

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A18
Heterogeneous Effects on Anxiety (GAD-7), India

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	0.168 (0.128)	0.244 (0.141)	0.255 (0.256)
Corporal Punishment $\times$ Male	0.102 (0.239)		
Corporal Punishment $\times$ Urban		-0.109 (0.315)	
Caregiver Education: Corporal Punishment $\times$ Any Education			-0.074 (0.314)
Observations	1774	1774	1774
R-squared	0.23	0.23	0.23

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A19
Heterogeneous Effects on Depression (PHQ-8), India

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	-0.080 (0.093)	0.113 (0.131)	-0.020 (0.228)
Corporal Punishment $\times$ Male	0.431* (0.209)		
Corporal Punishment $\times$ Urban		0.019 (0.216)	
Caregiver Education: Corporal Punishment $\times$ Any Education			0.254 (0.255)
Observations	1774	1774	1774
R-squared	0.28	0.28	0.28

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A20
Heterogeneous Effects on Education Enrollment, India

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	−0.004 (0.036)	0.008 (0.036)	−0.012 (0.052)
Corporal Punishment × Male	−0.010 (0.044)		
Corporal Punishment × Urban		−0.060 (0.047)	
Caregiver Education: Corporal Punishment × Any Education			0.006 (0.067)
Observations	1774	1774	1774
R-squared	0.21	0.21	0.21

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A21
Heterogeneous Effects on Employment, India

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Any Education
Corporal Punishment	0.030 (0.030)	0.034 (0.032)	0.026 (0.032)
Corporal Punishment × Male	0.008 (0.032)		
Corporal Punishment × Urban		−0.002 (0.045)	
Caregiver Education: Corporal Punishment × Any Education			0.014 (0.046)
Observations	1774	1774	1774
R-squared	0.38	0.37	0.38

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A22
Heterogeneous Effects on Cognitive Performance, Vietnam

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	-0.11* (0.06)	-0.13*** (0.04)	-0.11 (0.08)
Corporal Punishment × Male	0.03 (0.08)		
Corporal Punishment × Urban		0.15 (0.10)	
Caregiver Education: Corporal Punishment × Above Primary			0.04 (0.08)
Observations	1759	1759	1759
R-squared	0.40	0.41	0.40

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A23
Heterogeneous Effects on Psychosocial Performance, Vietnam

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	-0.20* (0.11)	-0.12 (0.07)	-0.25** (0.11)
Corporal Punishment × Male	0.08 (0.12)		
Corporal Punishment × Urban		-0.12 (0.16)	
Caregiver Education: Corporal Punishment × Above Primary			0.14 (0.12)
Observations	1758	1758	1758
R-squared	0.14	0.14	0.15

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A24
Heterogeneous Effects on Anxiety (GAD-7), Vietnam

	(1)	(2)	(3) Caregiver Education: Above Primary
	Male	Urban	
Corporal Punishment	0.11 (0.24)	0.05 (0.16)	-0.13 (0.23)
Corporal Punishment $\times$ Male	-0.14 (0.23)		
Corporal Punishment $\times$ Urban		-0.13 (0.25)	
Caregiver Education: Corporal Punishment $\times$ Above Primary			0.22 (0.19)
Observations	1553	1553	1553
R-squared	0.16	0.16	0.16

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A25
Heterogeneous Effects on Depression (PHQ-8), Vietnam

	(1)	(2)	(3) Caregiver Education: Above Primary
	Male	Urban	
Corporal Punishment	0.35 (0.23)	0.15 (0.12)	-0.12 (0.20)
Corporal Punishment $\times$ Male	-0.30 (0.28)		
Corporal Punishment $\times$ Urban		-0.02 (0.29)	
Caregiver Education: Corporal Punishment $\times$ Above Primary			0.42 (0.25)
Observations	1553	1553	1553
R-squared	0.19	0.19	0.19

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A26
Heterogeneous Effects on Education Enrollment, Vietnam

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	-0.11** (0.04)	-0.08 (0.05)	-0.06 (0.06)
Corporal Punishment $\times$ Male	0.05 (0.07)		
Corporal Punishment $\times$ Urban		0.01 (0.06)	
Caregiver Education: Corporal Punishment $\times$ Above Primary			-0.02 (0.06)
Observations	1553	1553	1553
R-squared	0.32	0.32	0.32

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A27
Heterogeneous Effects on Employment, Vietnam

	(1)	(2)	(3)
	Male	Urban	Caregiver Education: Above Primary
Corporal Punishment	0.07* (0.04)	0.06** (0.03)	0.11* (0.06)
Corporal Punishment $\times$ Male	-0.04 (0.05)		
Corporal Punishment $\times$ Urban		-0.06 (0.09)	
Caregiver Education: Corporal Punishment $\times$ Above Primary			-0.10 (0.07)
Observations	1553	1553	1553
R-squared	0.18	0.18	0.18

All regressions include controls and absorbed cluster FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$